When perception goes wrong

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Last time we looked at…

- **Bottom-up accounts** of visual perception
  - **Structuralism** (Wundt, Titchener)

    Method: **Description** of sensory elements

    Introspection, cataloguing

  - **Psychophysics** (Fechner, Weber, Stevens)

    Method: **Measurements** of sensory elements

    e.g., Methods of adjustment, limits, and constant stimuli

  - **Ecological theory** of perception (Gibson)

    Method: **Perception is offered** by the real world (object affordance)

    Invariant light information in the optic array

Last time we looked at...

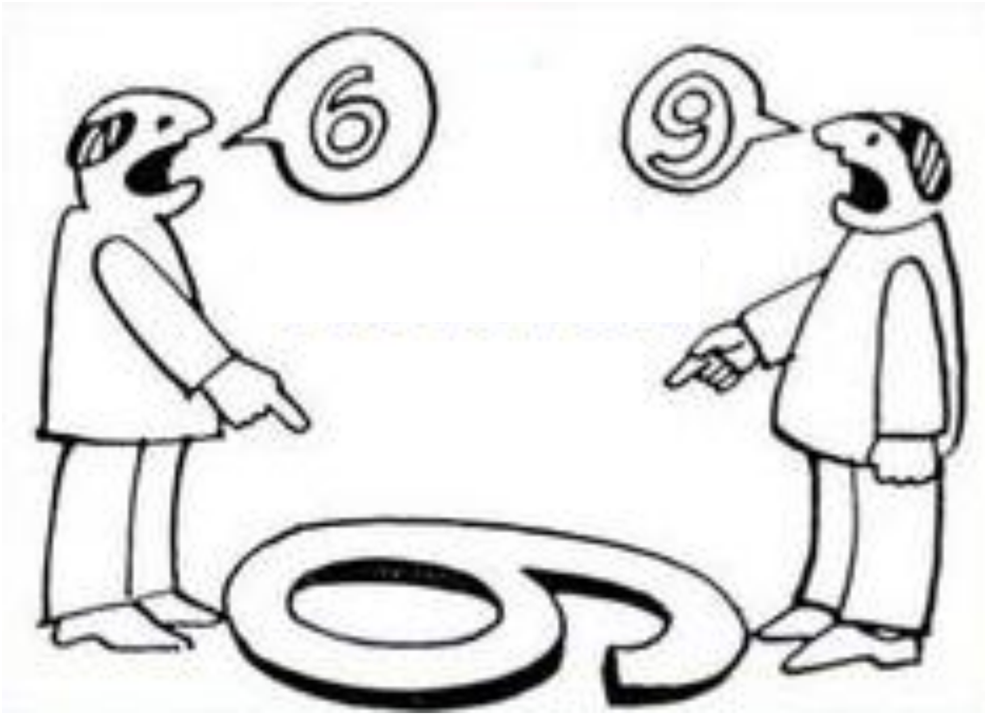
- **Bottom-up accounts** of visual perception
  - **Perception equals sensation**.
  - Perception is purely **stimulus-guided**.
  - Information gathered by the sensory receptors determines perception completely and automatically.

So, … perception is always accurate because it is directly based on the sensations provided by the real world.

…and the real world wouldn't lie to us... Would it?

Well, …

If perception equals sensation – who's right here?

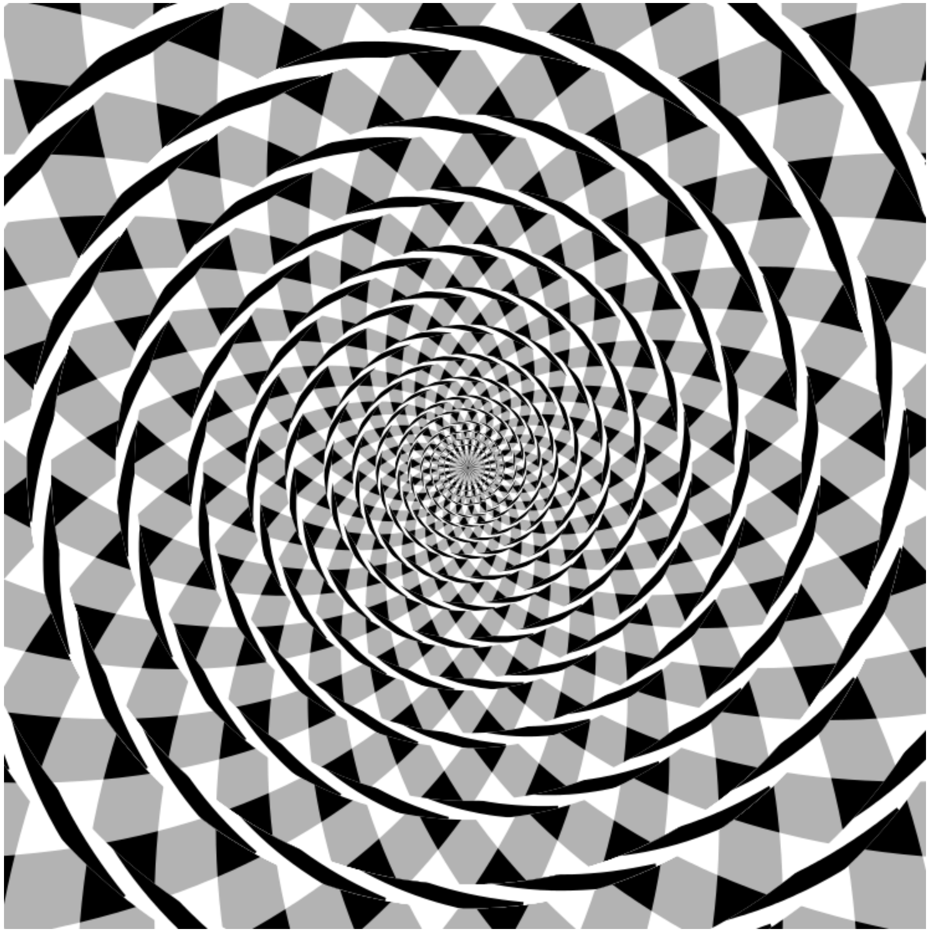


Erroneous perception - visual illusions

- Does the retinal image fully determine perception?
  - **Fraser spiral illusion**

    (James Fraser, 1908)

    What's the shape of the lines?



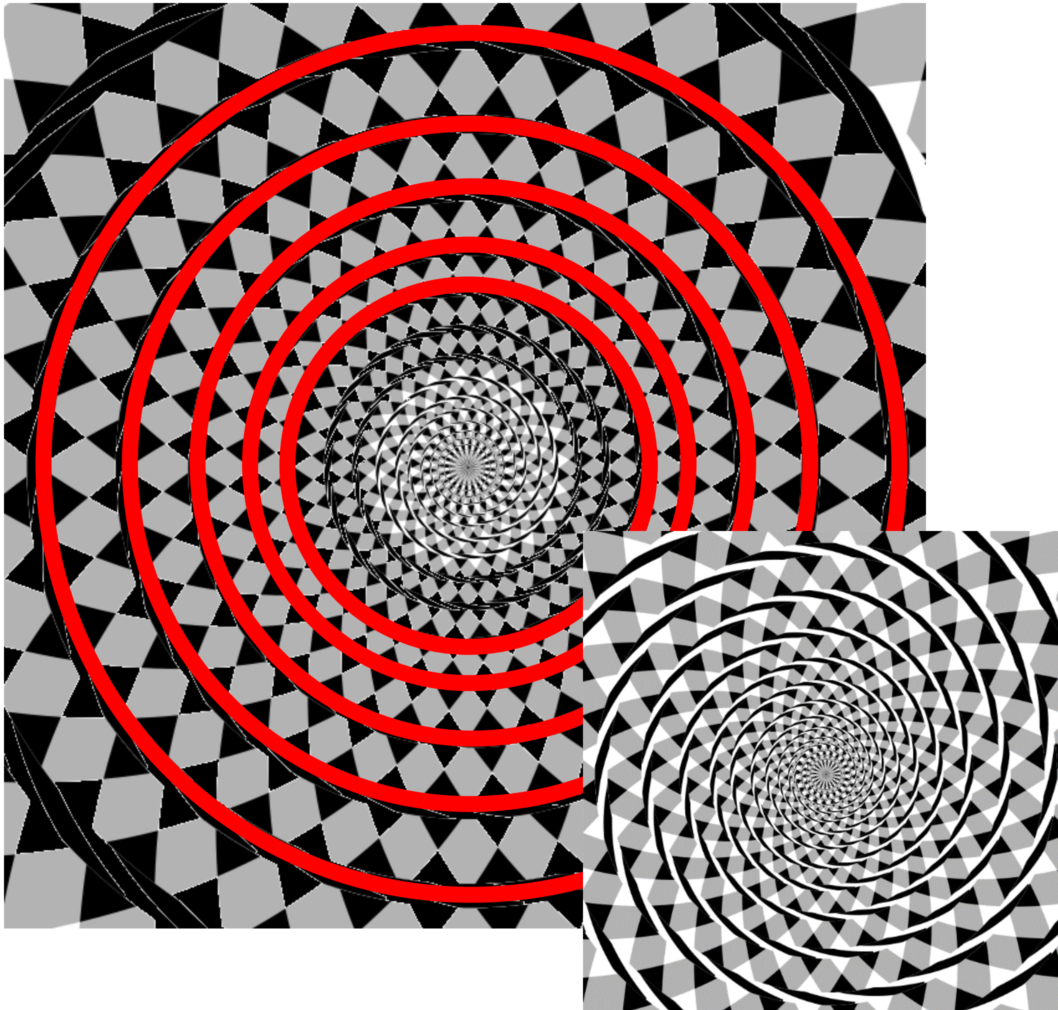
Erroneous perception - visual illusions

- Does the retinal image fully determine perception?
  - **Fraser spiral illusion**

    (James Fraser, 1908)

    What's the shape of the lines?

    Not a spiral, just a series of concentric circles.



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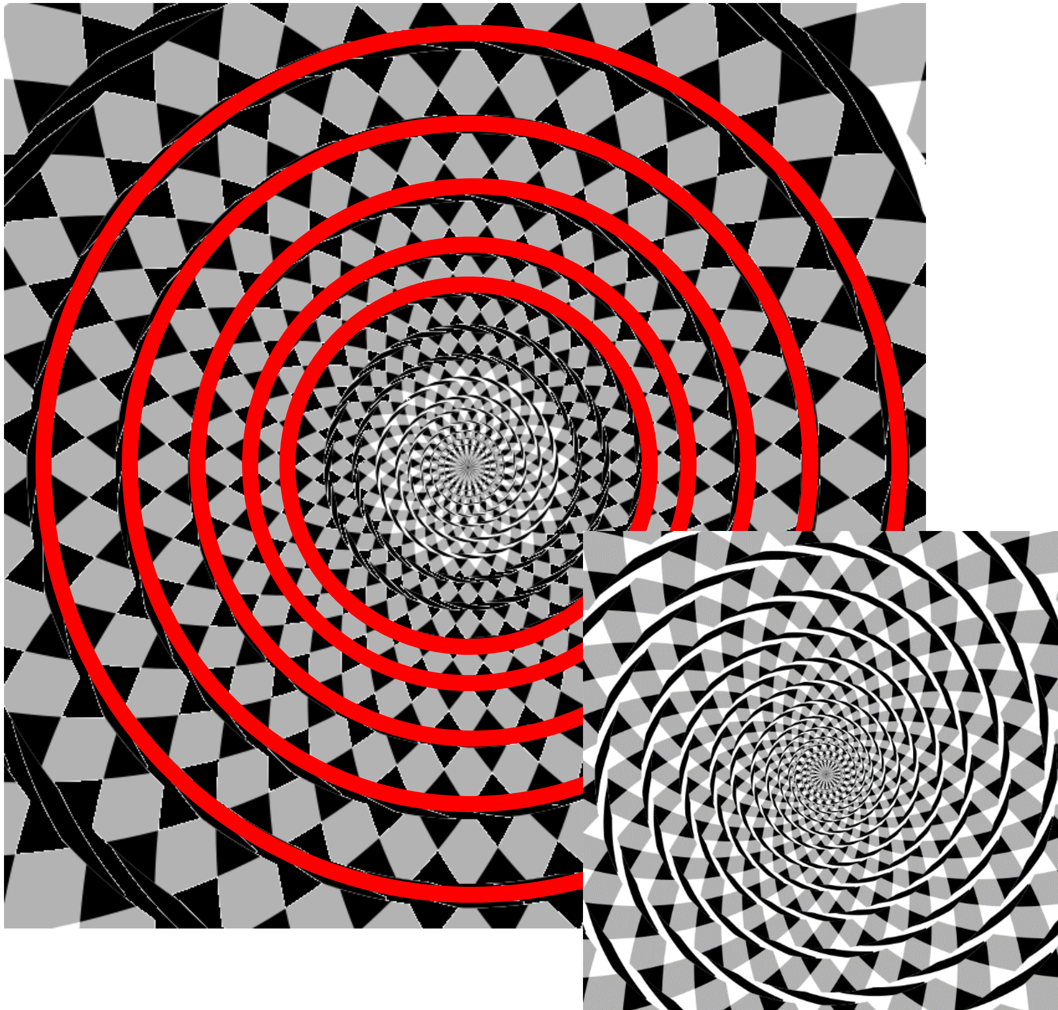
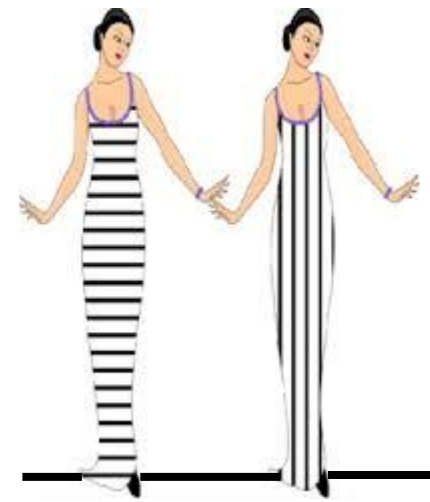


Misaligned parts (black and white strands within circles) distort the perception of the regular pattern (circles).

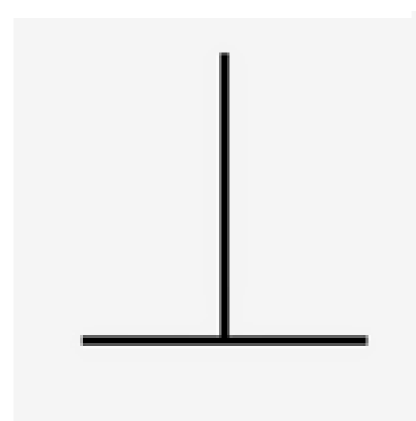
Erroneous perception - visual illusions

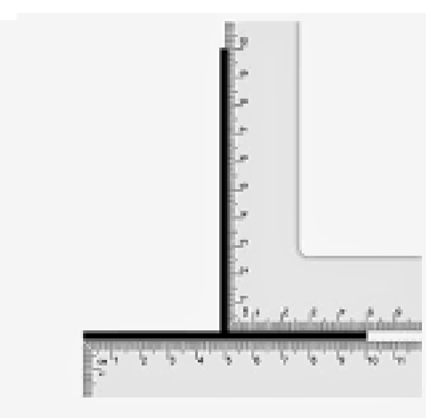
- Does the retinal image fully determine perception?
  - **Vertical-horizontal illusion**



Which line, horizontal or vertical, is longer?

They are equally long.





Bisecting lines are perceived as being longer than bisected (interrupted) lines.

Erroneous perception - visual illusions

- Does the retinal image fully determine perception?
  - **Jastrow illusion**

    (Joseph Jastrow, 1892)

Which bow is shorter?

They are equally long.

No confirmed explanation yet, but with a longer radius, objects appear to be shorter.

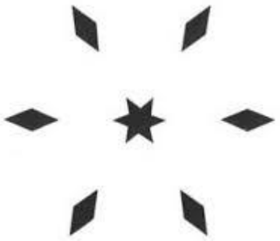
Erroneous perception - visual illusions

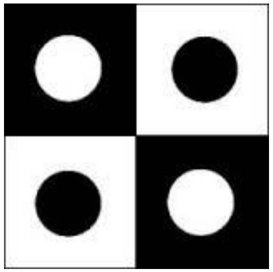
- Does the retinal image fully determine perception?
  - **Irradiation illusion**

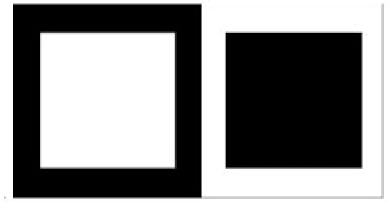
    (Hermann von Helmholtz, 1867)

    Which inner area, black or white, is larger?

    They have the same size.







Light areas appear to be larger than dark areas, because light from a white region irradiates adjacent dark regions.

Erroneous perception - visual illusions

- Does the retinal image fully determine perception?
  - **Irradiation illusion**

    (Hermann von Helmholtz, 1867)

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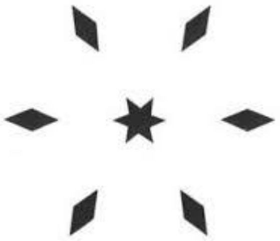
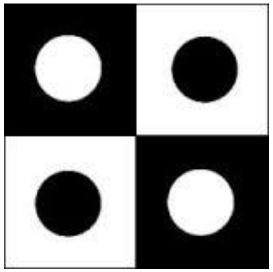
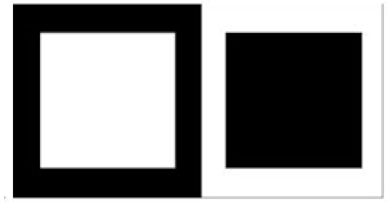
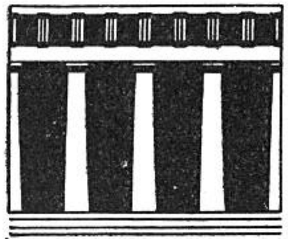
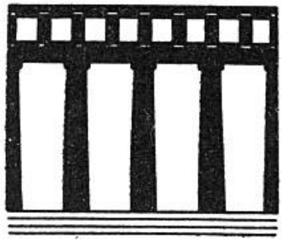








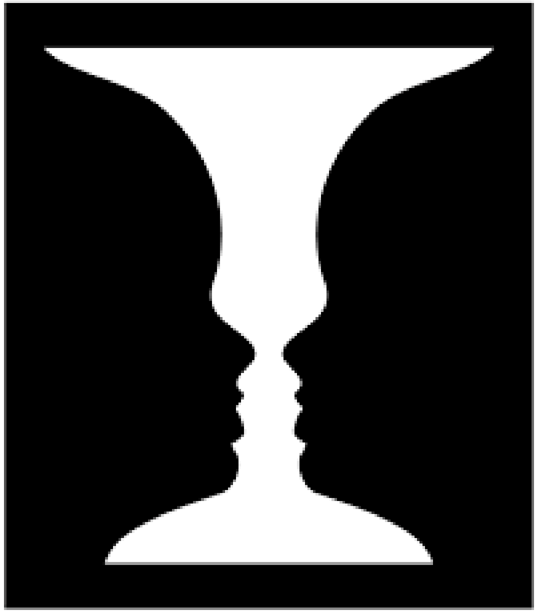


See how the Greeks adjusted their architecture (Parthenon) to compensate for optical illusions: http://www.architecturerevived.com/how-greek-temples-correct-visual-distortion/

Equivocal perception - ambiguous figures

- What if the image on the retina is ambiguous?
  - **Figure/Ground ambiguity**

    Vase or two faces?



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Equivocal perception - ambiguous figures

- What if the image on the retina is ambiguous?
  - **Figure/Ground ambiguity**

    Vase or two faces?

    The **retinal image doesn't change**, but we perceive different objects.

Equivocal perception - ambiguous figures

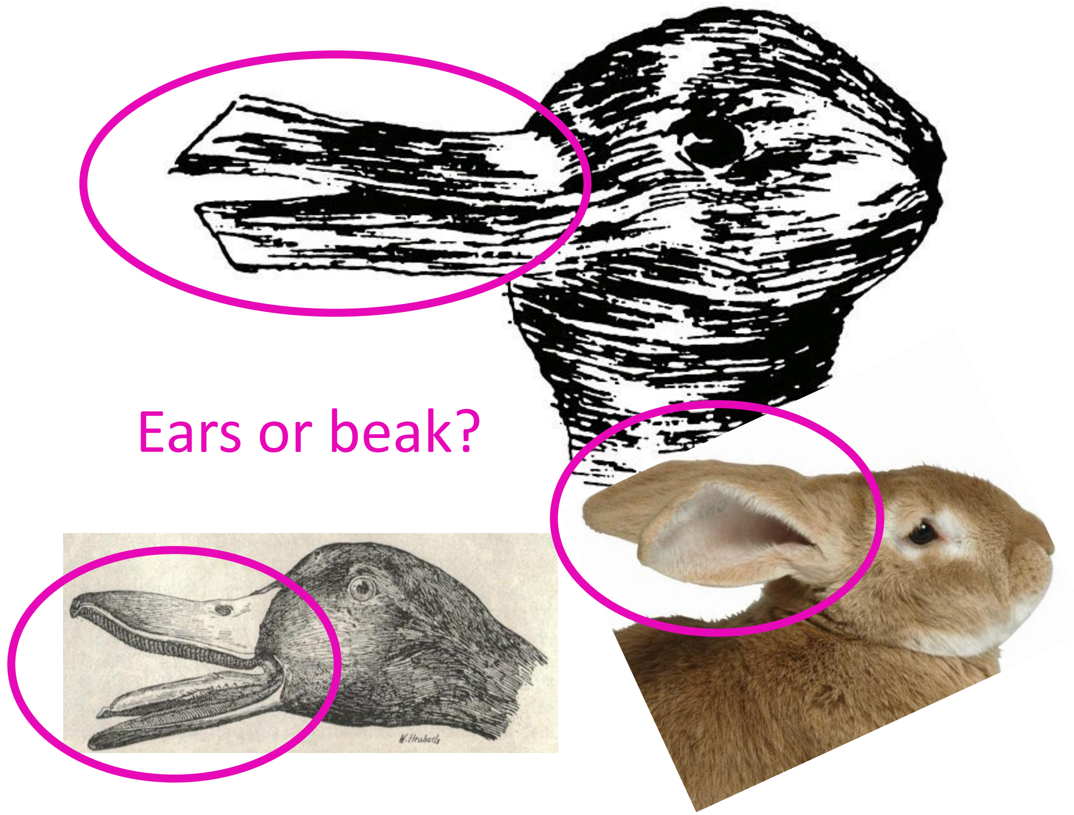
- What if the image on the retina is ambiguous?
  - **Figure/Ground ambiguity**

Focusing on the figure vs. the ground reveals **different objects**.

Equivocal perception - ambiguous figures

- What if the image on the retina is ambiguous?
  - **Feature ambiguity**

A duck or a rabbit?



Same feature can be **different** parts of different **objects**.

Equivocal perception - ambiguous figures

- What if the image on the retina is ambiguous?
  - **Feature ambiguity**

    Which head goes to which body?

Equivocal perception - ambiguous figures

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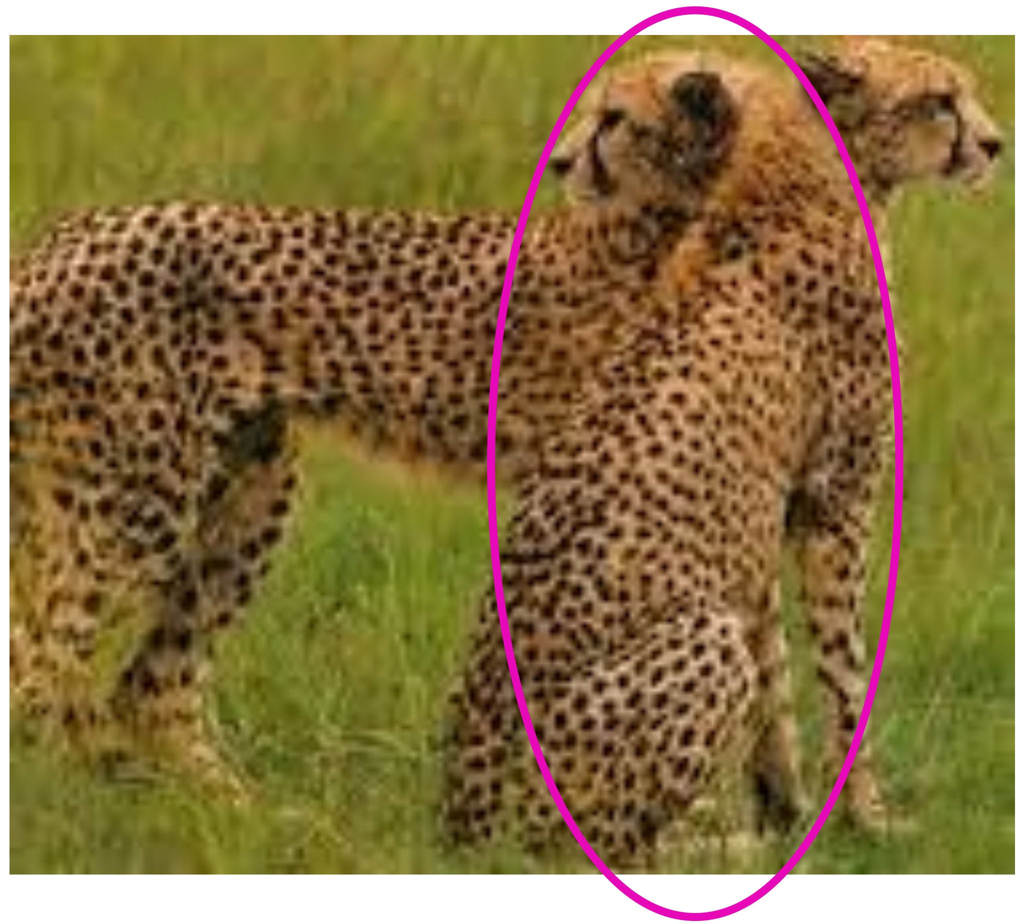

Equivocal perception - ambiguous figures

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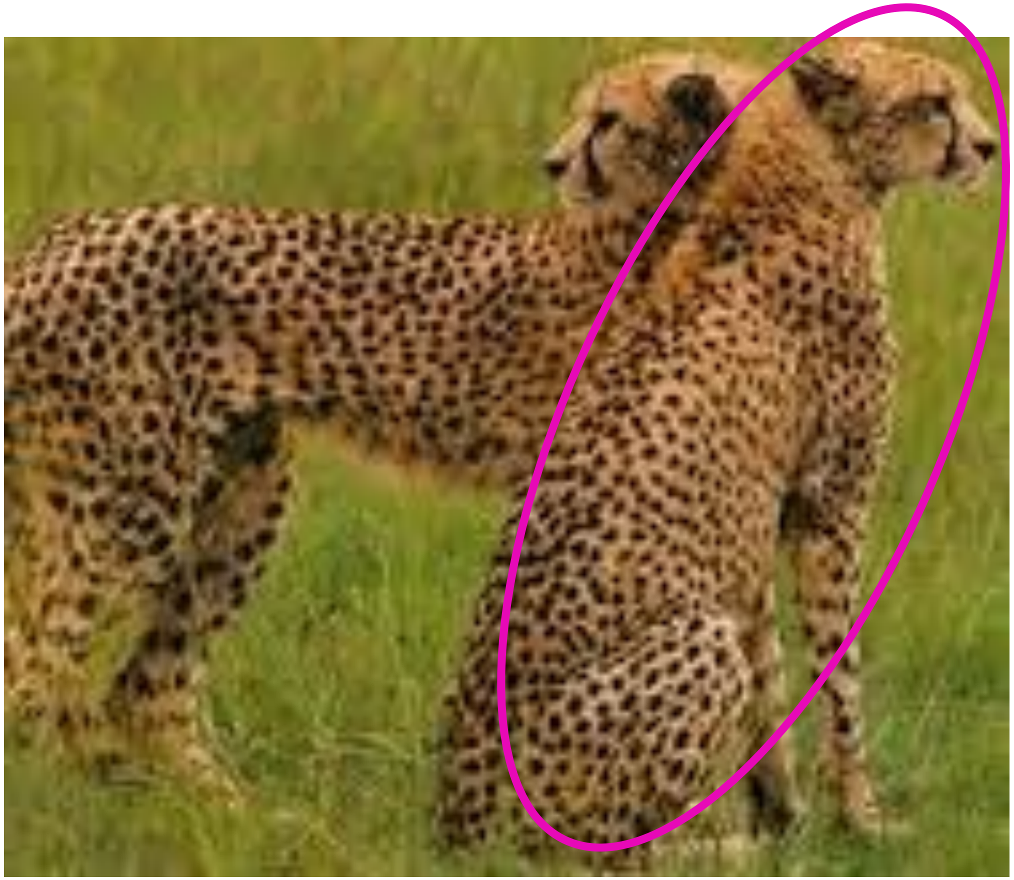


Equivocal perception - ambiguous figures

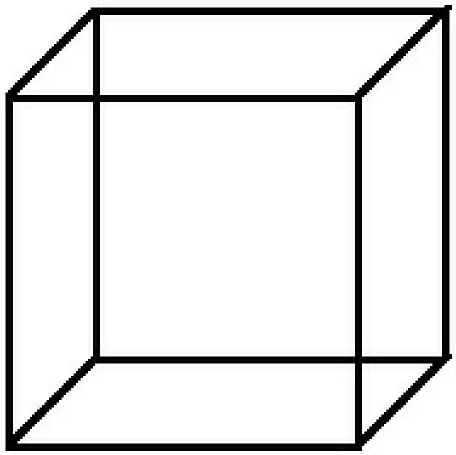
- What if the image on the retina is ambiguous?

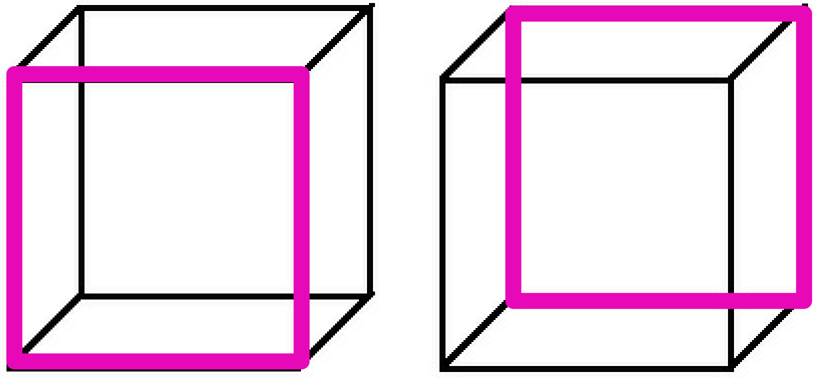
  - **Depth ambiguity**

    Necker cube

    (Louis Albert Necker, 1832)

    Where is the front
    and
    where the back?



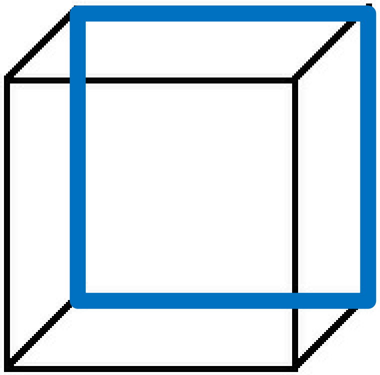


Equivocal perception - ambiguous figures

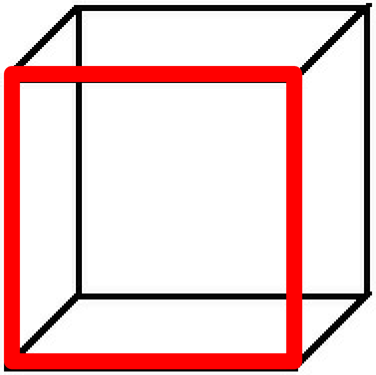
- What if the image on the retina is ambiguous?
  - **Depth ambiguity**

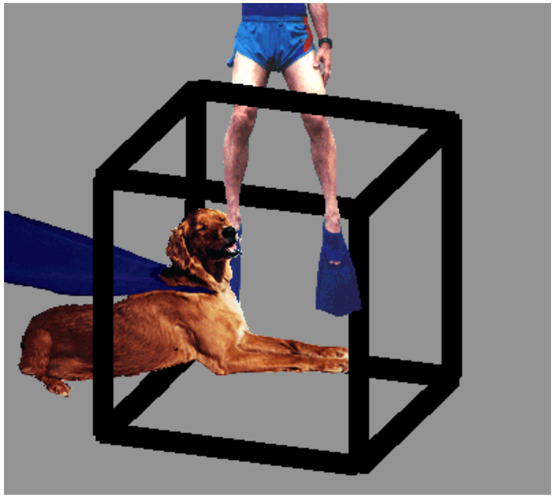
3D perception changes in the the **same retinal image**.

Focus on the legs; blue = front.

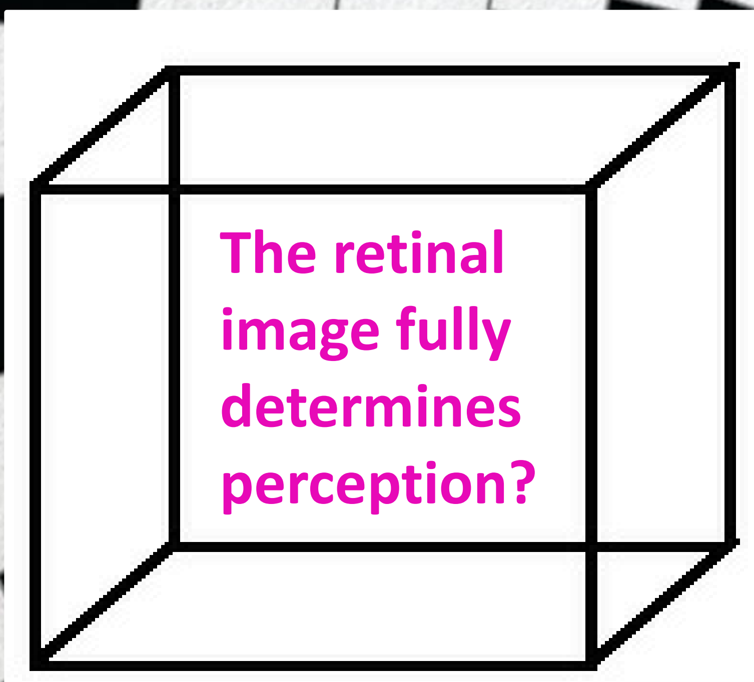


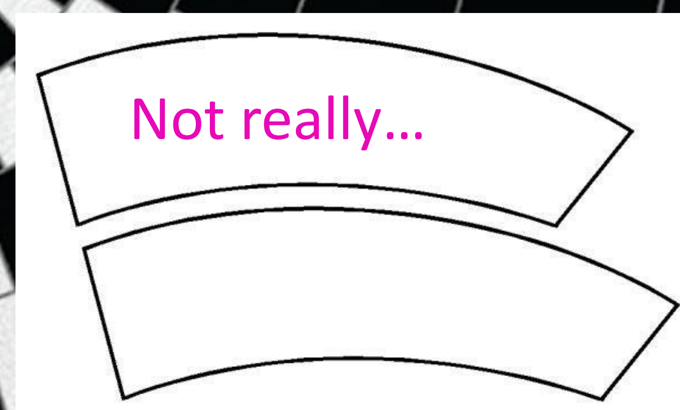
Focus on the dog; red = front.





- **Erroneous perception** with visual illusions.
  What we perceive is not always what is represented on our retina.
- **Equivocal perception** with ambiguous figures.
  We perceive different objects from the same retinal image.





- **Erroneous perception** with visual illusions.

  What we perceive is not always what is represented on our retina.

- **Equivocal perception** with ambiguous figures.

  We perceive different objects from the same retinal image.

Not really…

Can bottom-up accounts explain this?

Gibson? Any comments?

Such illusions and figures are carefully constructed to mislead us and do not exist in the real world. They have **no ecological validity**!

Real-world visual illusions

- **Moon illusion***

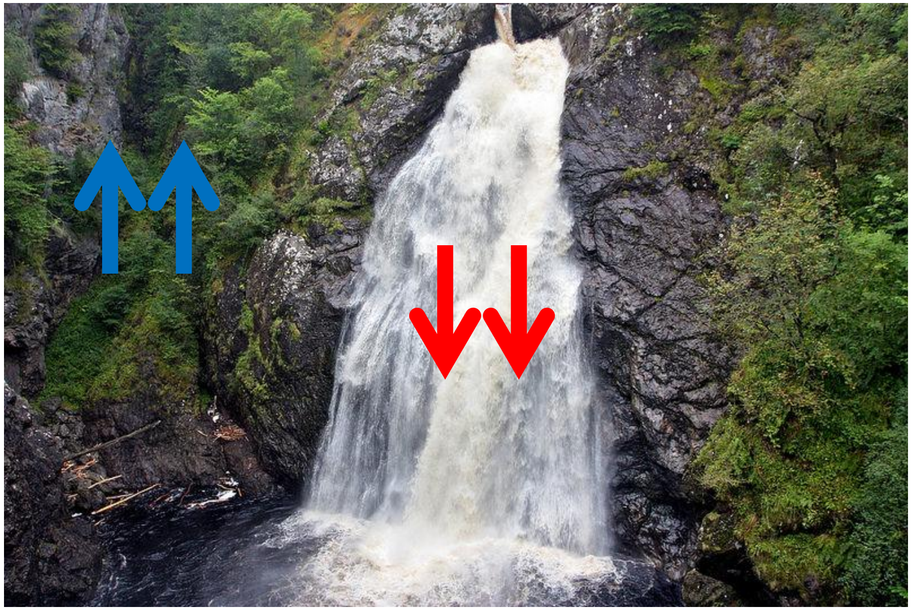
  * Explanation will follow in the size & depth perception lecture.

  – The moon appears larger when it is close to the horizon as compared to when it is high up in the sky.

Real-world visual illusions

- **Waterfall illusion**
  - **Motion aftereffect**: After observation of motion in one direction (waterfall for 30-60 seconds) stationary objects (e.g., trees) appear to move in the opposite direction.

Neuronal adaptation to motion



Explanation here: http://www.michaelbach.de/ot/mot-adapt/index.html

Real-world visual illusions

- **Wagon wheel effect**

Neurons try to make sense of fast motion of identical objects (blades of propeller) by interpolating frames, following the "nearest neighbour" principle.

  - **Stroboscopic effect**: A moving wheel appears to stand still or move in opposite direction to its true rotation.

Explanation here: http://www.michaelbach.de/ot/mot-wagonWheel/index.html

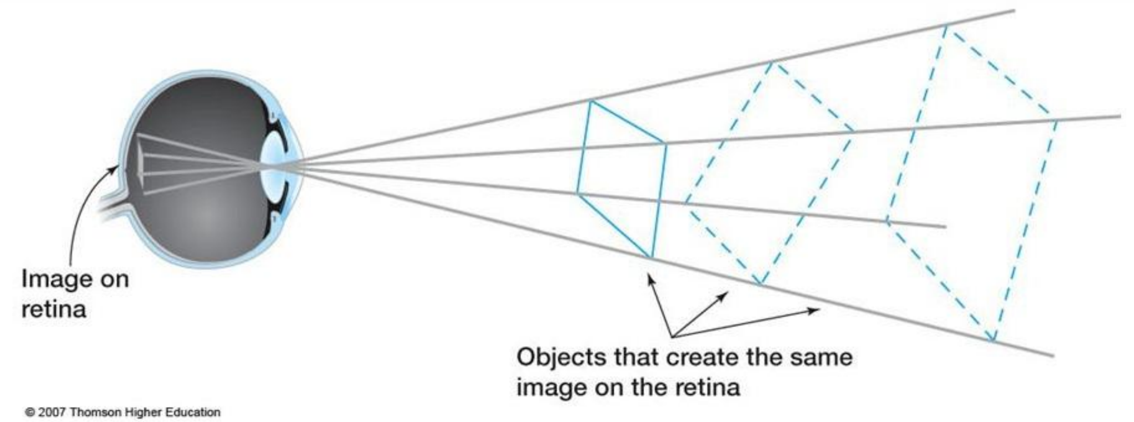
Real-world visual ambiguity

- Ambiguous retinal images
  - The **inverse projection problem**

    A 3D object is represented on a 2D (retinal) surface.

    The same pattern of light on the retina can be caused by different objects.

    The real-world 3D object cannot be derived from the retinal image.



Real-world visual ambiguity

- Ambiguous retinal images
  - **Superimposed objects**
  - **Untypical angles**

Theoretical accounts of perception (2)

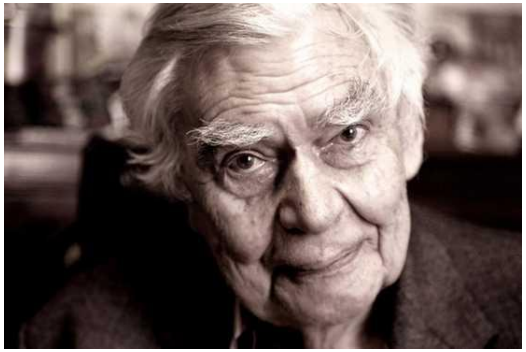
- The fact that we experience **perceptual illusions** and **ambiguity** tells us that…
  - Perception is more than just sensation.

    Sensory information from the retina is insufficient, **perception is a matter of interpretation**.

- Theories accounting for this are
  - **Top-down accounts**

    Perception is an interaction between **sensation and cognition**, i.e., stored object knowledge, context information, personal expectation and motivation, are used to help interpreting sensory stimulation.

Theoretical accounts of perception (2)



- **Richard Gregory**, 1970
  - **Constructive theory** of perception
  - Basic idea

    Perception is **indirect** (interpreted), and a **constructive** process of **hypothesis testing** (hypotheses, based on internal factors are used to interpret the sensory input).

    For ambiguous figures two equally plausible hypotheses are established.

    As perception is based on **individual factors**, incorrect hypotheses can be formed, which lead to **perceptual errors**.

    As for visual illusions.

Theoretical accounts of perception (2)

- Constructive theory of perception – summary
  - Goal

    Explain how we attach **meaning** to sensory input.

  - Perception = **Interpretation** of sensation.

    Coffee mug = The sensory input (round, cup-shaped, smell, …) matches the perceptual hypothesis (based on stored object representations, situational context, personal expectations and motivation, …).

Theoretical accounts of perception (2)

- Constructive theory of perception – summary
  - Evaluation

    (+) **Theoretical account** (tries to explain how perception and object recognition work).

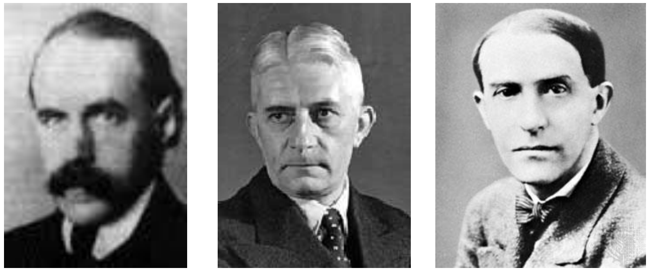
    (+) Is able to explain **perceptual failure** (visual illusion) and **ambiguity**.

    (-) Cannot explain **why illusions persist** even when they are known.

    (-) Suggests that perception is **not effortless** (active and continuous hypothesis testing).

    Does not really match our subjective experience.

Theoretical accounts of perception (3)

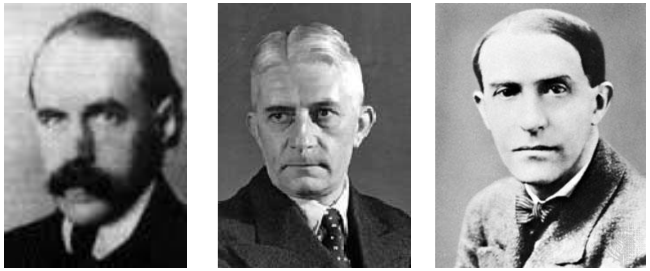


- Gestalt psychologist
  - Max **Wertheimer**/Wolfgang **Köhler**/Kurt **Koffka**, 1910

Two light flashes presented in a rapid alternating fashion create illusory movement.

Modern version of the “phi phenomenon”: Apparent motion induced by sequentially presented static stimuli

Theoretical accounts of perception (3)



- Gestalt psychologist
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Two light flashes presented in a rapid alternating fashion create illusory movement.

The sensory experience (two light flashes) is not sufficient to explain the perceptual experience (illusion of movement).

“The whole is more than the sum of its parts“

Theoretical accounts of perception (3)

- Gestalt (i.e., the whole configuration)
  - Basic question
    How do we achieve the whole (i.e., the object)?
  - Answer
    By **perceptual organisation**, such as…
    …grouping (putting together elements that form an object).
    …segmentation (separating different groups of elements).

    Remember:
    Ganglion cells have centre-surround receptive fields that are designed to detect luminance discontinuities.
    V4 neurons respond to (colour or texture) discontinuities enabling us to detect that some areas in the visual field have **similar** properties while others **differ** from each other.

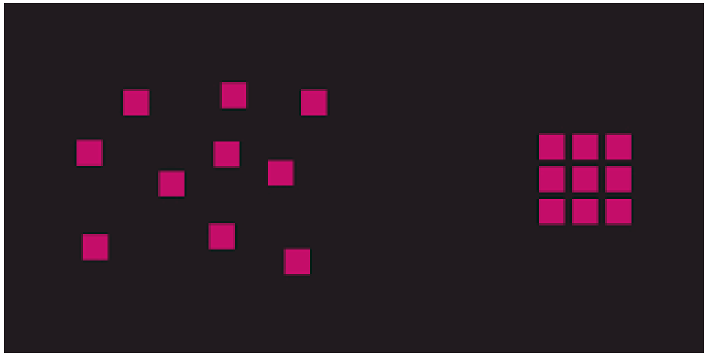
Theoretical accounts of perception (3)

- Perceptual organisation
  - **Principles** (heuristics) that lead to whole.
    1. Proximity
    2. Similarity
    3. Common fait
    4. Good continuation
    5. Closure
    6. Relative size
    7. Surroundedness
    8. Orientation
    9. Symmetry

Mnemonic anyone?
Please share, if you do.

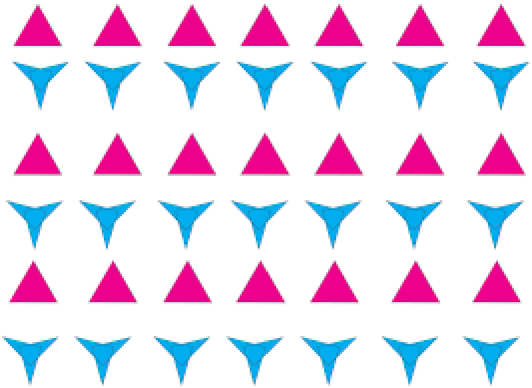
Gestalt principles of perceptual organisation

- Proximity
  - Elements that are **close together** are grouped together.



- Similarity
  - Elements that **look similar** are grouped together.

Similarity seems to override proximity.



Gestalt principles of perceptual organisation

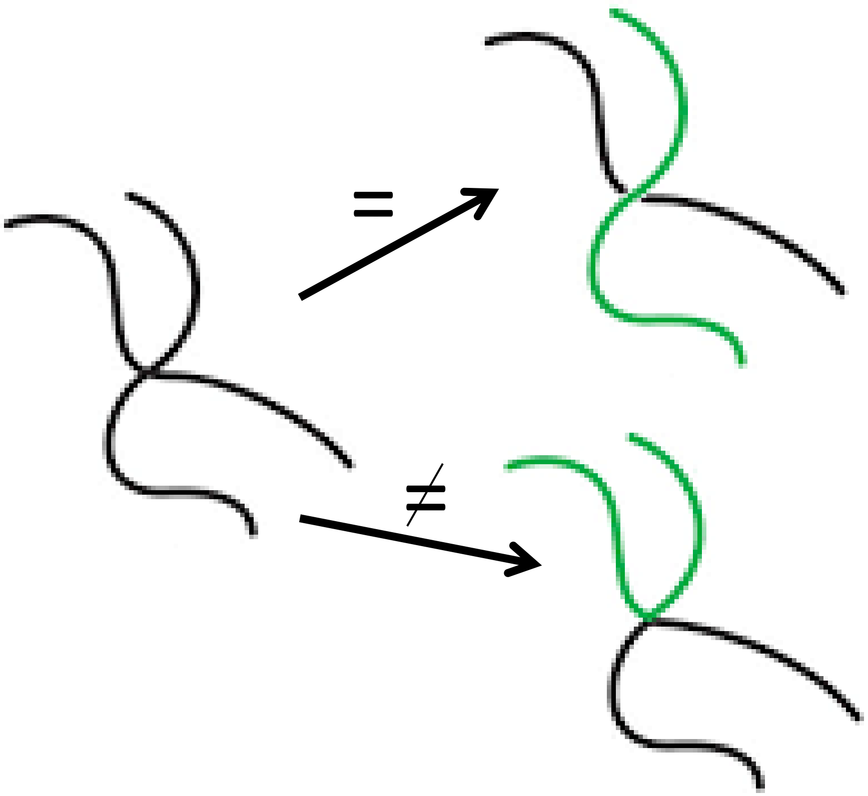
- Common fate
  - Elements that (appear to) **move together** are grouped together.

This is why camouflage animals can be seen when they move (their parts are grouped together).

Gestalt principles of perceptual organisation

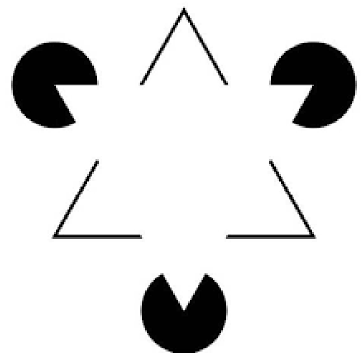
- Good continuation
  - Elements that **continue** are grouped together.

    Static version of common fate.



Gestalt principles of perceptual organisation

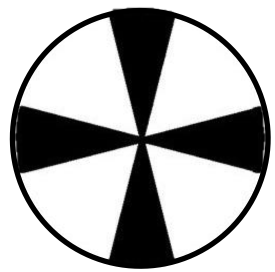
- Closure
  - Elements that **close a figure** are grouped together.

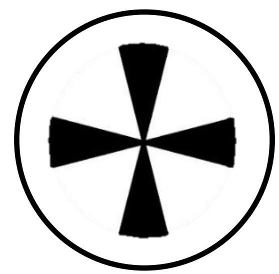


Gestalt principles of perceptual organisation

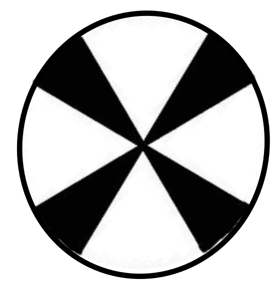
- Relative size, surroundedness, orientation, and symmetry
  - Elements that are

    Relatively **smaller**

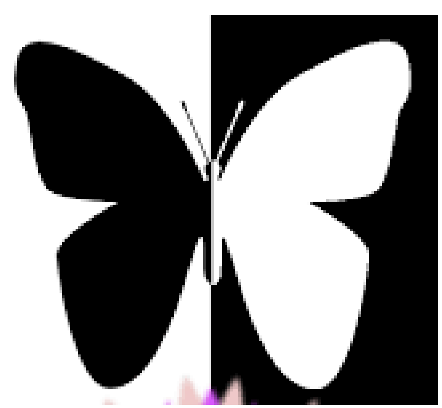




    In a **surrounded** area



    Horizontally/vertically **oriented**



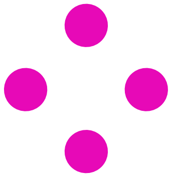
    Symmetrical

  are grouped together.

The Gestalt law of perceptual organisation

- **Law of Prägnanz** (good Gestalt)

  "Of several geometrically possible organisations that one will actually occur which possesses the best, simplest, and most stable shape."

  e.g.,  will be perceived as a square because it matches the shape of square more than the shape of a triangle plus an extra dot.

Theoretical accounts of perception (3)

- Gestalt psychology – summary
  - Goal

    Explain how we attach **meaning** to sensory input.

  - Perception = **Interpretation** of sensation in line with the law of Prägnanz.

    Coffee mug = The sensory elements (round, cup-shaped, handle, …) are grouped together in a meaningful way (based on the principles of perceptual organisation).

Theoretical accounts of perception (3)

- Gestalt psychology – summary
  - Evaluation

    (+) Provides a set of useful **perceptual heuristics** (principles of how sensory input is grouped and segmented).

    (-) Aims to be a theoretical account but is **mainly descriptive** (describes what grouping principles are, not how object perception and recognition work).

Theoretical accounts of perception (4)

For an attempt to **integrate top-down and bottom-up accounts** look at Neisser, U. (1976). *Cognition and reality: Principles and implications of cognitive psychology*. New York: Freeman.

For an **integration of ecological and constructivist accounts** in terms of dorsal and ventral processing streams, read: Norman, J. (2002). Two visual systems and two theories of perception: An attempt to reconcile the constructivist and ecological approaches. *Behavioural and Brain Sciences, 25*, 73-144.

Theoretical accounts of perception (5)

A quite different, but very interesting approach to visual perception is **David Marr's (1982) computational theory of visual perception**. If you want to know more continue reading Rookes & Wilson (p. 34-37). If you want to know much more, read his latest book: Marr, D. (2010). Vision: A Computational Investigation into the Human Representation and Processing of Visual Information. Cambridge: MIT Press.

More of this in the object **perception & recognition lecture**.

Finally…

- Perception can go wrong on many levels.
  - There are **more than 200 visual illusions** based on…

    colour, luminance, contrast, motion, geometry, size, shape, …

    We will meet some more of them!

If you cannot wait, look at **Prof Michael Bach**'s tremendously cool homepage to find an excessive list of visual illusions, with animations, explanations and everything!
While you're there – watch the mouse cursor!

http://www.michaelbach.de/ot/

Summary

- We have seen that perception can be erroneous and ambiguous…

  … and that last week's **bottom-up accounts fail** to explain this,…

  … while top-down accounts can.

- We have looked at **2 top-down accounts** of perception.

  1) Gregory's **constructive theory** of perception

     Sensory input is interpreted by means of perceptual hypotheses (cognition).

  2) **Gestalt psychology**

     Sensory input is grouped and segmented (perceptually organised) following principles of the good Gestalt.

Test yourself – learning goals

- After reviewing this lecture, you should be able to
  - Name and explain some visual illusions and ambiguities.
  - Define what the inverse projection problem is.
  - Compare bottom-up and top-down theories of perception.
  - Discuss why bottom-up accounts cannot, but top-down accounts can explain visual illusions and the existence of ambiguous figures.
  - Explain what the Gestalt psychologists mean with perceptual organisation and how it is achieved.
  - Name and explain the Gestalt principles.

  - **Brownie point**: If you find a good mnemonic sentence to remember the Gestalt principles, please let me know!

Test yourself – multiple choice

1. Which description does not fit Gregory's constructive approach of perception?

   a) Stimulus-led.

   b) Top-down controlled.

   c) Knowledge-guided.

   d) Concept-driven.

Test yourself – multiple choice

2. The Gestalt psychologist claimed that 'the whole is more than the sum of its parts'. This statement puts them in the corner of theories advocating that _______________

a) Perception equals sensation.

b) Perception is automatic.

c) Perception is sensory summation.

d) Perception is interpreted sensation.

Test yourself – multiple choice

3. The Gestalt principle of closure is associated with which of these stimulus properties?

   a) Continuation.

   b) Shape.

   c) Segmentation.

   d) Size.

See you next week…